

Last time: Many results for $\text{IPS}/_{\mathbb{R}}$ carry over to $\text{IPS}/_{\mathbb{C}}$. Examples include $\sqrt{T}$ and polar decomposition.

Theorem (SVD) Let $V, W : \text{IPS}/_{\mathbb{C}}, \dim V = m, \dim W = n (m, n < \infty)$. Then $\exists$ ONB $v_1, \dots, v_m \in V, w_1, \dots, w_n \in W, \sigma_1 \geq \dots \geq \sigma_p \geq 0, p := \min\{m, n\}$ s.t. $Tv_i = \sigma_i w_i$.

The sequence σ_i is uniquely determined by T , called the singular values of T .

Proof $T^*T \in \text{End}(V)$ is self-adjoint, positive semi-definite. From Spectral Theorem we claim its eigenvalues $\lambda_1 \geq \dots \geq \lambda_m \geq 0$, with corresponding ONB $v_1, \dots, v_m$. Take $\sigma_i := \sqrt{\lambda_i}$, and rest is same as the real version. $\square$

In terms of matrices, let $V = \mathbb{C}^m, W = \mathbb{C}^n$ + std IP, T corresponds to $A \in M_{m \times n}(\mathbb{C})$. Then $\exists$ unitary matrix $P \in M_{m \times m}(\mathbb{C}), Q \in M_{n \times n}(\mathbb{C})$ s.t. ${}^tQAP = \text{diag}(\sigma_1, \dots, \sigma_p, 0, \dots, 0)$, i.e. $A = Q \text{diag}(\sigma_1, \dots, \sigma_p, 0, \dots, 0) {}^tP$.

In fact, $P = (v_1 | \dots | v_m), Q = (w_1 | \dots | w_n)$ in terms of ONB.

Definition-Proposition (Moore-Penrose inverse) $\forall T \in \text{Hom}(V, W), \exists ! S \in \text{Hom}(W, V)$ s.t.

1. $TST = S$,
2. $STS = T$,
3. $(ST)^* = ST$,
4. $(TS)^* = TS$.

Proof Same as the real case. $\square$

The connection between SVD and MP inverse remains. If $\{v_i\}, \{w_j\}, \{\sigma_k\}$ are as in SVD of T , then

$$Sw_j = \begin{cases} \sigma_j^{-1} v_j, & j \leq \text{rk} T \\ 0, & j > \text{rk} T \end{cases}$$

$$\text{In term of matrices, } A = Q \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{pmatrix} {}^tP \implies \text{MP inverse} = P \begin{pmatrix} \sigma_1^{-1} & & \\ & \ddots & \\ & & \sigma_r^{-1} \end{pmatrix} {}^tQ.$$

That is all for complex inner product spaces. Continue with real cases.

§ Orthogonal transformations / operators on $\text{IPS}/_{\mathbb{R}}$

In this section we will classify all orthogonal operators on $\text{IPS}/_{\mathbb{R}}$ by showing that they share similar formats. We shall start by recalling some basic knowledge.

Let $V : \text{IPS}/_{\mathbb{R}}, \dim < \infty$. Recall that T is orthogonal if $T^* = T^{-1}$ (equivalently: T is linear, invertible and preserves inner product). In the case of $V = \mathbb{R}^n$ + std IP, orthogonal matrix $A \in M_{n \times n}(\mathbb{R})$ satisfies ${}^tA = A^{-1}$.

Observation T orthogonal $\implies \det T = \pm 1 \because A {}^tA = 1 \implies (\det A)^2 = 1$.

Definition We say $T \in \text{End}(V)$ is **normal** if $T^*T = TT^*$. Note that T may be non-diagonalizable, so Spectral Theorem is unavailable.

Lemma If T is normal and $\exists k \geq 0$ s.t. $T^k = 0 \in \text{End}(V)$, then $T = 0_V \in \text{End}(V)$.

Proof Work in matrices. Assume $T \leftrightarrow A \in M_{n \times n}(\mathbb{R}), {}^tAA = A {}^tA$. Since $\mathbb{R} \subset \mathbb{C}, A$ is also normal in the sense of complex IPS. Spectral Theorem tells us $\exists P : \text{unitary}, P^{-1}AP =$

$$\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}, \lambda_i \in \mathbb{C}. \text{ Therefore } A^k = P^{-1} \begin{pmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{pmatrix} P = 0 \implies \forall i, \lambda_i = 0. \quad \square$$

Next we will classify orthogonal operators on $V : \text{IPS}/_{\mathbb{R}}$.

Observation $A \in M_{n \times n}(\mathbb{R}) : \text{orthogonal} \iff A = (v_1 | \dots | v_n), \{v_i\} : \text{ONB of } V.$

- $n = 1 : \text{End}(V) \simeq \mathbb{R}$, i.e. orthogonal operators $= \{\pm \text{id}_V\}$.
- $n = 2 : \text{Assume } V = \mathbb{R}^2 + \text{std IP}.$

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \text{ is orthogonal} \iff \begin{cases} \alpha^2 + \gamma^2 = \beta^2 + \delta^2 = 1 \\ \alpha\beta + \gamma\delta = 0 \end{cases}$$

Take $\theta \in \mathbb{R}$ s.t. $\alpha = \cos \theta, \gamma = \sin \theta$. All unit vectors $\perp (\alpha, \gamma)$ are

$$\{(\sin \theta, -\cos \theta), (-\sin \theta, \cos \theta)\} \ni (\beta, \delta),$$

with $\pm$ determined by $\begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix} \in \{\pm 1\}$. Specifically

$$\begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} = 1, \quad \begin{vmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{vmatrix} = -1.$$

Remark Fixing $\theta \in 2\pi\mathbb{Z}$ and $\det = 1$ gives $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, which acts on $\mathbb{R}$ by reflection on x -axis.

When $\det = 1$, we define

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} =: R(\theta),$$

called the **rotation matrix** of angle $\theta \bmod 2\pi$.

- $R(0) = 1_{2 \times 2}, \quad R(-1) = -1_{2 \times 2}.$
- $R(\theta_1) = R(\theta_2) \iff \theta_1 = \theta_2 \pmod{2\pi}.$
- $R(\theta)R(\varphi) = R(\theta + \varphi) = R(\varphi)R(\theta) \implies \text{commutative}.$
- $R(\theta) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} \implies R(\theta) \begin{pmatrix} \cos \varphi \\ \sin \varphi \end{pmatrix} = R(\theta)R(\varphi) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos(\theta + \varphi) \\ \sin(\theta + \varphi) \end{pmatrix}.$

Lemma Let $\theta \in \mathbb{R}, P \in M_{2 \times 2}(\mathbb{R}) : \text{orthogonal}, \varepsilon := \det P \in \{\pm 1\}$. Then

$$P^{-1}R(\theta)P = R(\varepsilon\theta).$$

Proof

- $\varepsilon = 1 : \exists \varphi \in \mathbb{R}, P = R(\varphi)$. Then $P^{-1}R(\theta)P = R(-\varphi + \theta + \varphi) = R(\varepsilon\theta).$
- $\varepsilon = -1 : \text{can write } P = R(\varphi) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$ Then

$$P^{-1}R(\theta)P = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} R(-\varphi)R(\theta)R(\varphi) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} R(\theta) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = R(-\theta).$$

□

Theorem Let $n := \dim V, T \in \text{End}(V) : \text{orthogonal}.$ $\exists$ ONB under which T becomes

$$\text{diag}(1_a, -1_b, R(\theta_1), \dots, R(\theta_m))$$

where $a + b + 2m = n, \quad \theta_i \in \mathbb{R} \setminus \mathbb{Z}\pi.$

Lemma If $T : \text{orthogonal}$, then $T + T^{-1}$ is self-adjoint with eigenvalues $\in [-2, 2].$

Proof $(T + T^{-1})^* = T^* + T^{**} = T + T^*$.

Assume $V = \mathbb{R}^n$, T viewed as $n \times n$ matrix, orthogonal. Since $\mathbb{R} \subset \mathbb{C}$ and $T^*T = TT^*$, from Spectral Theorem $\exists P$: unitary, $P^{-1}TP = \text{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_i \in \mathbb{C}$, $|\lambda_i| = 1$.
 $\implies P^{-1}(T + T^{-1})P = \text{diag}(\lambda_1 + \lambda_1^{-1}, \dots, \lambda_n + \lambda_n^{-1}) \implies |\lambda_i + \lambda_i^{-1}| \leq 2$. $\square$

Proof of Theorem Let $S = T + T^{-1}$, then $V = \bigoplus_{\lambda \in [-2, 2]} V_\lambda$ (orthogonal). Note that $ST = T^2 + 1 = TS \implies V_\lambda$ is T -invariant. Hence reduce to the case $S = \lambda \cdot \text{id}$, $\lambda \in [-2, 2]$.

- If $\lambda \in \{\pm 2\}$, then $\underbrace{(T \pm \text{id})^2}_{\text{normal}} = 0_V \xrightarrow{\text{Lem}} T = \mp \text{id}$.
- Assume $|\lambda| < 2$, then $X^2 - \lambda X + 1 = 0$ has no real roots (irreducible on $\mathbb{R}$) $\implies$ It is Min_T .
 As a result, T has no real eigenvalues.

Now fix $v \in V$, $v \neq 0$, then v, Tv are linearly independent. Let $W = \langle v, Tv \rangle$, $\dim W = 2$.
 From $T^2 - \lambda T + \text{id} = 0_V$ one can show W is T -invariant, hence $W^\perp$ is $(T^* = T^{-1})$ -invariant.
 Also $T = \lambda \cdot \text{id} - T^{-1}$ gives $T = f(T^{-1})$, so $W^\perp$ is T -invariant as well with $\dim W^\perp = n - 2$.

Reduce to the case $V = V_\lambda$, $|\lambda| < 2$, $\dim V = 2$ via induction on $\dim V$ and replacing V with W . By comparing dimension, $X^2 - \lambda X + 1 = 0$ is exactly Char_T . From $\det T = 1$, $T = R(\theta)$ under some θ and ONB. Here $\theta \notin \pi\mathbb{Z}$, otherwise $S = T + T^{-1}$ has eigenvalue $\in \{\pm 2\}$. $\square$

Remark

- The data $a, b, \theta_1, \dots, \theta_m$ are uniquely determined by T up to permutation and $\pm\theta_i + 2k\pi$.
- a = the multiplicity of 1 as a root of Char_A . b and θ_i is similar.
- $\det T = (-1)^b$.

Next time: Special case $\dim V = 3$.

Definition When $\dim V = 3$, $V : \text{IPS}/\mathbb{R}$, we say $T \in \text{End}(V)$ is a **rotation** if T is orthogonal and $\det T = 1$. From Thm we know

$$T \xleftrightarrow{\exists \text{ ONB}} \begin{pmatrix} 1 & & \\ & R(\theta) & \\ & & 1 \end{pmatrix} \text{ or } \begin{pmatrix} 1 & & \\ & -1 & \\ & & -1 \end{pmatrix}.$$